CivicConnect

Project Engineering Document v3.0

Construction, verification and release evidence index

REVIEW CANDIDATE - 16 OCTOBER 2026
Release decision: NOT READY for final baseline/public production release.
Prepared for Emile, Don and Masego. Final candidate approval and formal decision reconciliation remain pending.

This continues the repository PED: M1 sections 01-15 and M2 sections 16/17/19/20/21 remain the historical sources. This export indexes those sources and adds section 22; historical sign-offs retain their original scope.

Local candidate evidence: 50 automated tests passed; two live HTTP journeys; p95 0.422 seconds over five minutes with 10 users/500 requests; local restart and backup/restore passed. Evidence boundaries and remaining gaps follow.

22. M3 construction, verification and release review

CivicConnect - PED v3.0 review candidate, 16 October 2026. Prepared by Emile with AI assistance recorded in AI-036. Continues the existing M1/M2 PED sources; no historical signature or baseline status is changed. No new student approval is recorded.

22.1 Candidate and construction

The integration candidate combines the current main snapshot with the proposed backend, route tests, UI and dependency-check changes in PRs #80-84. Source revisions and file hashes are indexed in `docs/release/evidence/2026-10-16/source-manifest.json`. The registration addition is identified separately by `registration-source-manifest.json`. A final signed release SHA/tag is pending.

Implemented slice: public Requester registration, login, submission, own list/detail/status; Staff login, queue/status filter/detail/acceptance; acceptance notification through in-app and SMTP; Manager/Administrator login with a safe landing page. `docs/deployment/m3-interface-agreement.md` records actual routes, forms, roles and error behaviour. `CHG-002` records the proposed release-slice exclusions and impact without rewriting baselined priorities.

Two integration defects were corrected: missing CSRF tokens in shared logout forms on list/landing pages (DEF-004), and a mismatched audit-actor field in the detail template (DEF-005). Both have regression checks.

22.2 Architecture and configuration conformance

The candidate follows layered server-rendered Flask, service-level rules, SQLite persistence, transactional state/audit/outbox changes and an in-process notification worker. ADR-002/003/004/006 remain Accepted in the source; ADR-001/007 remain Proposed. DEC-008 remains Deferred. The six Issue #65 register rows are draft proposals awaiting team reconciliation, not accepted decisions.

Staging is a distinct loopback Waitress process with a generated synthetic database, separate credentials and local STARTTLS SMTP sink. Config/secrets and the database stay Git-ignored. This simulates staging and does not establish public HTTPS, availability or production durability. See `docs/deployment/m3-staging.md`.

22.3 Verification results

Evidence	Result	Scope
pytest automated suite	50 passed in 75.07s	registration-regression.xml; 11 new registration cases; earlier 39-case snapshot retained
Backend regression follow-up	16 passed	Includes corrected rendered audit actor assertion
Ruff / Bandit / dependency audit	Clean / no medium-high findings / no known application dependency findings	Tool results at check time, not complete security proof
Live HTTP system journey 1	Passed	Requester login, submit #501, receipt, list/detail, logout and cross-requester denial
Live HTTP system journey 2	Passed	Staff filter/detail/accept, requester status, in-app notification and local SMTP capture
Browser walkthrough	Request #502 submitted and accepted	Actual browser UI operated; form/receipt screenshots retained; no checked-in browser automation claim
NFR-001 five-minute load	8,503 successful; 0 failures; p95 0.422s	10 distinct Staff users, 500 filtered records, local warm instance, target <=2s
Local process restart	Passed	Accepted #501 retrieved after stop/start of same-file staging
Online backup/separate restore	Passed	Integrity check OK; request/audit/outbox counts preserved

`docs/release/evidence/2026-10-16/` holds execution reports. The five-second smoke run is explicitly insufficient for NFR-001. The subsequent 300-second run supplies the stated measurement. The load/restart/backup evidence predates registration and was not rerun. Registration-live.json separately confirms public signup and login; email verification and signup rate limiting remain outstanding. No browser rendering benchmark or production-capacity claim is made.

22.4 Traceability and quality controls

`docs/requirements/RTM.csv` adds an execution-evidence column for the implemented slice without promoting baseline statuses. `docs/quality/test-inventory.md` remains the historical kickoff; `docs/quality/m3-verification-handoff.md` records the new boundary and remaining independent checks. Requirements, acceptance criteria, code, named tests, execution reports, DEF-004/005 and release conditions are connected through these artefacts.

GitHub workflows provide automated tests, static analysis, dependency audit and secret-pattern checks. Local results are not GitHub CI results. Current CI and branch-protection enforcement must be checked on the signed final revision. Two separate non-author students must review it; two accounts belonging to one person do not create two reviews.

22.5 Residual risk and release decision

NOT READY for final baseline/public production release. The scoped local demonstration candidate has passing evidence, while independent reviews, formal scope/decision reconciliation and the final committed revision remain pending. See the full release summary for conditions and owners.

Unresolved claims include durable hosted storage, external SMTP/TLS, dormant legacy-hash disposition, complete role x operation coverage, stuck/duplicate outbox recovery, reversible schema/code rollback, sustained availability, cold-start, capacity and accessibility. A local backup drill does not prove daily backup scheduling or retention. Local restart success does not close the M2 Render redeploy failure.

Render documents that free-service local files disappear on restart/redeploy/spindown, which is consistent with the observed M2 failure. The historical instance configuration was not independently recovered. Avoiding redeploy alone cannot ensure durability. PostgreSQL remains a candidate, not an adopted platform.

22.6 Evidence index and individual defence

Engineering concern	Authoritative artefact
M1 business/scope/requirements baseline	PED sections 01-15 and docs/requirements/ registers
M2 architecture/data/patterns/integration/baseline	PED sections 16, 17, 19, 20, 21 and docs/decisions/adr/
Current route/UI contract	docs/deployment/m3-interface-agreement.md
Controlled staging and recovery	docs/deployment/m3-staging.md; scripts/m3_staging.py; scripts/m3_verify_recovery.py
Test classifications and remaining matrix	docs/quality/m3-verification-handoff.md; tests/
Performance scenario and result	performance/load_test_staff_queue.py; performance.json
Live journeys and restart evidence	scripts/m3_verify_live.py; live-system-journeys.json; process-restart.json
Traceability and defects	docs/requirements/RTM.csv; docs/quality/defect-register.csv
Debt and residual risks	docs/quality/technical-debt-register.md; docs/release/m3-release-evidence-summary.md
Decisions/scope requiring review	docs/release/team-decision-review.md; docs/decisions/issue-65-draft-rows.csv; CHG-002
AI accountability and progressive history	docs/governance/ai-usage-register.csv; PROJECT_HISTORY.md; final PR/commit

Each student should be able to demonstrate: FR-014/AC-014 -> service version guard -> acceptance regression -> live journey #501 -> audit/outbox/notification evidence -> residual outbox recovery risk -> release condition. Emile should also explain NFR-001's dataset, concurrency, warm-up, duration, p95 method and local-only limits.

22.7 Final review actions

1. Emile inspects this candidate. The authenticated GitHub commit records technical authorship; personal acceptance and final baseline signatures remain separate.

2. Don reviews backend/migration/outbox, existing ADR statuses and the six draft register rows.

3. Masego independently verifies test classification, role/privacy coverage and RTM; both non-author reviewers review the exact integration diff.

4. Record actual scope/decision agreement, resolve findings, pass required CI and merge through the PR process.

5. Record the final SHA/tag, actual approvals and residual-risk acceptance; regenerate the controlled PED v3 PDF. This review export is not the final signed baseline.

Sources checked 16 October 2026: https://flask.palletsprojects.com/en/stable/testing/ and https://render.com/docs/free. Repository artefacts above supply implementation and execution evidence.